How Gambling Behaviors Change Among Below-Median Income Individuals After Pandemics (Signals of "Poor Tax")

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1 Introduction and Motivation

Australians lose more to gambling per capita than any other country in the world. Gambling is often not simply a matter of individual entertainment - it has macroeconomic consequences. Public health campaigns in Australia reflect widespread recognition that gambling contributes to social harms, from addiction to household financial stress. Beyond the individual-level consequences of gambling behavior, a particularly concerning secondary effect is its potential to exacerbate wealth inequality - essentially acting as a "poor" tax (disproportionately affecting lower income groups). The COVID-19 pandemic introduced a unique disruption in people's incomes, leisure, as well as limited access to physical venues, forcing a prompt shift to online gambling. We endeavour to answer: Did gambling continue to act as a regressive "poor tax" during the pandemic, or did the unique conditions and limited access reduce income-based differences in gambling behaviour?

2 Data and Methods

2.1 Data Source

The data source is provided by Roy Morgan for education purposes with confidentiality. The assigned subset covers consumption information for individuals aged 14 and above in New South Wales.

2.2 Key Data Manipulation

2.2.1 COVID-19 Timeline

According to the NSW Ministry of Health (2023), the dataset is divided into 5 subsets with respect to the sample period:

- Pre-COVID: Jan - Feb 2020
- Early-COVID: Mar 2020 - May 2021
- Lockdown: Jun - Oct 2021

- Post Restrictions: Nov 2021 - Dec 2022
- Living With COVID Jan 2023 - Dec 2024

2.2.2 Outcome Variables

There are nine boolean values describing whether an individual has ever participated in that gambling activity (i.e. an observation takes 1 if an individual has participated in that activity, and 0 otherwise). These gambling activities are Casino, Racing, Poker Machines, TAB, Scratch Tickets, Lotteries, Poker, Internet Gambling and Bookmakers. We used them as outcome variables.

2.2.3 Income Level Classifications

As per the Australian Bureau of Statistics (2024), the income levels (in AUD) are separated as follows:

- Below the Median: $\leq$ 50000-59999
- Above the Median: >59999 and $\leq$ 150000 - 159999
- High Income > 159999

We are particularly interested in the group of individuals with incomes below the median.

2.2.4 Other Data Cleaning

The variable "full time job", containing more than 40000 missing values, was removed. Individuals under the age of 18 have been removed as gambling is age-restricted. We also classified individuals' employment status, marital status, education level, religions and life stages.

2.3 Methods

We estimated for each of the 5 COVID periods and 9 gambling activities using models such as the following:

$$\text{Models}_{m \times n} = \text{Periods}_m \times \text{GamblingTypes}_n$$

for m = 1,...,5, n = 1,...,9. Categorical variables were converted into dummy variables using one-hot encoding, to avoid the dummy variable trap.

2.3.1 Logistic Regression with LASSO Regularization and Cross-Validation

After standardization, logistic regression is used to explain results in a probabilistic man- ner, while systematically avoiding probability axiom violation. L1 regularization is introduced to discover solid linear relationships between the outcome variable and features. It penalizes and excludes irrelevant features, therefore improving parsimony. Cross-validation is used to simulate out-of-sample performance, and select the optimal penalty term producing the best in-sample fitness.

2.3.2 Random Forest

We also used random forest for complementary comparison. With two distinct modeling philosophies, such comparison addresses whether augmenting complex non-linearities (Random Forest) outperforms the regularized approach (LASSO).

2.3.3 Methods Comparison-Parametric Regularized Linear model vs. Non-Parametric Ensemble Method

Table 1: Average Performance Comparison between LASSO and Random Forest Models. See appendices for formulas

Metric	**LASSO**	**Random Forest**	**Difference**	**Better Model**
AME Calibration Error	0.0093	0.1918	0.1825	LASSO
AUC	0.9468	0.9684	0.0216	Random Forest
Brier Score	0.0405	0.0963	0.0558	LASSO

The above table displays average performance metrics across all 45 models. The AME Calibration Error indicates that Random Forest systematically overestimated probabilities while LASSO produces a fairer result. This is because each decision tree in a Random Forest sees bootstrap samples during training. Due to the limited sample size and scarcity of positive examples, the trees tend to assign higher probabilities "conservatively" when uncertain, to avoid missing positive examples. Similar to calibration error, Brier Score also

shows that LASSO produces more accurate results than Random Forest. For AUC scores, both methods perform similarly, showing fair discrimination.

2.3.4 Average Marginal Effect

We chose to use the Average Marginal Effect (AME) for interpretation. Unlike log-odds, AME constrains probabilities to a more interpretable 0-1 scale. AME also standardizes compositional differences such as demographic shifts across periods by stating: "The predicted probabilities of the entire population having below-median incomes in each period". Since individuals' income levels may change over time, such an interpretation method allows for temporal comparisons.

3 Results and Interpretations

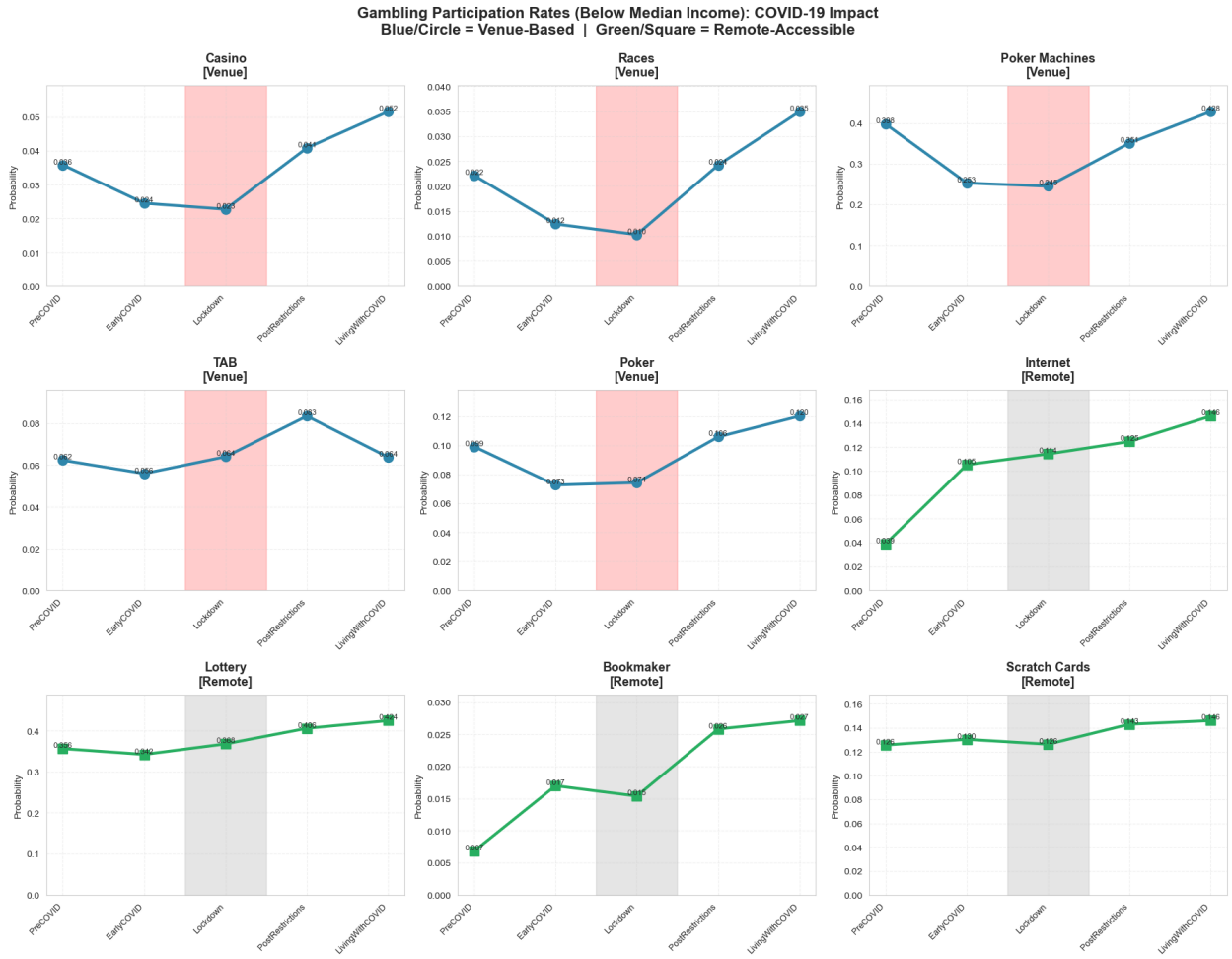

Figure 1: Predicted Probabilities (AME) across all five period and nine gambling activities. Each data point is a predicted probability

The distinct trajectories reveal how COVID differentially impacted gambling modes. Venue-based activities requiring physical presence declined during lockdown as expected, but notably recovered in the post-COVID era, suggesting pent-up demand. Online gambling's threefold increase demonstrates accelerated digital adoption. Suggesting below-median income individuals shifted toward remote-accessible and lower-threshold gambling since COVID. These results illustrate that less wealthy people's demand for gambling increased and reveal signals of "poor tax". As per the NSW government: "You win some, you lose more".

4 Conclusion and Limitations

The increasing demand among eight out of nine gambling activities suggests below-median income individuals tend to gamble more post-pandemic. Also, shifting to online gambling may expose them to greater risk. Given the data collected is self-reported, respondents may answer "Yes" to some gambling types such as lotteries, bookmakers, racing, and internet gambling simultaneously. Since these gambling activities can occur both online and offline, the corresponding predicted probabilities may not be accurate. Moreover, we assume that the majority of individuals are risk-averse, and that their risk preferences remain constant throughout the sample period.

5 Further Extensions

Due to insufficient evidence, our current results cannot be interpreted as causal. However, causality could be argued by further controlling potential confounders such as how individuals' willingness to gamble varies with their wealth/disposable income. With this, we can also potentially explore how COVID reshapes individuals' consumption patterns across income levels.

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Appendices

Table 2: AME Comparison Results (First 7 Columns)

Gambling Type	Period	True Rate (Low)	LASSO AME	RF AME	LASSO Calib. Error	RF Calib. Error
CASIN_CASIN_TOTL	PreCOVID	0.028951	0.035811	0.055787	0.006859	0.026836
CASIN_CASIN_TOTL	EarlyCOVID	0.017736	0.024470	0.201983	0.006733	0.184247
CASIN_CASIN_TOTL	Lockdown	0.015588	0.022698	0.185010	0.007110	0.169422
CASIN_CASIN_TOTL	PostRestrictions	0.027392	0.040841	0.301756	0.013449	0.274364
CASIN_CASIN_TOTL	LivingWithCOVID	0.035389	0.051572	0.331794	0.016183	0.296405
RACES_RACES_TOTL	PreCOVID	0.021127	0.022139	0.040486	0.001012	0.019359
RACES_RACES_TOTL	EarlyCOVID	0.008929	0.012407	0.169884	0.003478	0.160955
RACES_RACES_TOTL	Lockdown	0.006141	0.010280	0.102216	0.004139	0.096075
RACES_RACES_TOTL	PostRestrictions	0.017791	0.024206	0.308065	0.006415	0.290274
RACES_RACES_TOTL	LivingWithCOVID	0.026730	0.034916	0.342949	0.008186	0.316219
POKERM_POKERM_TOTL	PreCOVID	0.394366	0.397616	0.444449	0.003250	0.050083
POKERM_POKERM_TOTL	EarlyCOVID	0.252534	0.252623	0.427296	0.000089	0.174762
POKERM_POKERM_TOTL	Lockdown	0.244450	0.244782	0.396519	0.000333	0.152069
POKERM_POKERM_TOTL	PostRestrictions	0.333199	0.350607	0.460983	0.017408	0.127784
POKERM_POKERM_TOTL	LivingWithCOVID	0.399385	0.427748	0.480598	0.028363	0.081213
TAB_TAB_TOTL	PreCOVID	0.067293	0.062325	0.175628	0.004968	0.108335
TAB_TAB_TOTL	EarlyCOVID	0.046091	0.055980	0.326764	0.009889	0.280673
TAB_TAB_TOTL	Lockdown	0.041332	0.064002	0.288329	0.022670	0.246997
TAB_TAB_TOTL	PostRestrictions	0.067808	0.083451	0.380919	0.015643	0.313111
TAB_TAB_TOTL	LivingWithCOVID	0.049445	0.063759	0.357358	0.014314	0.307913

AUC Score Formula in General Case

$$AUC = \int_0^1 TPR(x)d(FPR(x))$$

- TPR = $\frac{TP}{TP+FN}$ (True Positive Rate)
- FPR = $\frac{FP}{FP+TN}$ (False Positive Rate)

Average Calibration Error Formula

$$Average\ Calibration\ Error = \frac{1}{n}\sum_{i=1}^{n}(x_i^{measured} - x_i^{true})$$

Brier Score

$$BrierScore = \frac{1}{N}\sum_{i=1}^{N}(f_i - o_i)^2$$

- N = total number of samples
- f_i = predicted probability that the event will occur (model output)
- o_i = actual outcome (1 if the event occurred, 0 if it did not

Logistic Lasso Model Specification

For a binary outcome $y_i \in \{0, 1\}$ and feature vector $x_i \in \mathbb{R}^p$, the logistic model is:

$$\begin{aligned} P(y_i = 1|x_i) &= \sigma(\beta_0 + \beta^\top x_i) \\ &= \frac{1}{1 + \exp(-(\beta_0 + \beta^\top x_i))} \end{aligned}$$

where $\sigma(z) = (1 + \exp(-z))^{-1}$ is the logistic function.

Lasso Regularization

The Logistic Lasso estimates parameters by solving the regularized optimization problem:

$$\min_{\beta_0, \beta} \left\{ -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] + \lambda \|\beta\|_1 \right\} \tag{1}$$

Code and Data Link (OneDrive)

https://unimelbcloud-my.sharepoint.com/:f:/g/personal/kuoyuanw_student_unimelb_edu_au/EtBNZBn1HBdLgZRiqt8GLtcBGY-EojsQhfcjO_sXO74mRw?e=DTfN5b